Classical Ciphers

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POLYALPHABETIC CIPHER

Definition:

Polyalphabetic ciphers use multiple substitution alphabets to encrypt data, making it more secure than monoalphabetic ciphers.

Why Polyalphabetic?

Unlike monoalphabetic ciphers, they prevent frequency analysis attacks by spreading letter substitutions over different alphabets.

POLYALPHABETIC CIPHER

Working

Basic Concept: In a polyalphabetic cipher, each letter in the plaintext is shifted differently depending on its position in the text.

Cipher Text Generation: Involves multiple Caesar ciphers applied to different parts of the message.

Key Structure: The cipher is controlled by a keyword, determining which alphabet is used.

Introduction to Vigenère Cipher

Definition:
The Vigenère cipher is a method of encrypting alphabetic text by using a series of different Caesar ciphers based on the letters of a keyword.

History:

- Named after **Blaise de Vigenère**, though it was **Giovanni Battista Bellaso** who first proposed the cipher in 1553.
- Popularized in the 19th century for military and diplomatic use.

Key Features of Vigenère Cipher

Polyalphabetic Substitution Cipher:
Uses multiple Caesar ciphers based on a repeating keyword.

Key:
The key can be any string of letters, and it repeats throughout the plaintext.

Prevention of Frequency Analysis:
Vigenère ciphers obscure letter frequency patterns by using different substitution alphabets, making it harder to break.

Encryption Process

Plaintext: The original message that needs to be encrypted.

Key: A word or phrase that controls the shifts in each letter of the plaintext.

Encryption Formula:

$$Ci=(Pi+Ki) \bmod 26$$

where:

Pi is the index of the plaintext character.
Ki is the index of the corresponding key character.
Ci is the resulting ciphertext character.

Step-by-Step Encryption Example

Plaintext: "ATTACKATDAWN"
Key: "LEMON" (repeated as "LEMONLEMONLE")

1.Shift Calculations:

1. A (0) + L (11) = 11 → L
2. T (19) + E (4) = 23 → X
3. T (19) + M (12) = 31 (mod 26) → F
4. A (0) + O (14) = 14 → O
5. C (2) + N (13) = 15 → P
6. K (10) + L (11) = 21 → V

A	B	C	D	E	F	G	H
0	1	2	3	4	5	6	7
I	J	K	L	M	N	O	P
8	9	10	11	12	13	14	15
Q	R	S	T	U	V	W	X
16	17	18	19	20	21	22	23
			Y	Z			
			24	25			

2.Resulting Ciphertext:

LXFOPVEFRNHR

Decryption Process

Decryption Formula:

$$Pi=(Ci-Ki) \bmod 26$$

To reverse the encryption, subtract the key values from the ciphertext.

Step-by-Step Decryption Example:

Ciphertext: "LXFOPVEFRNHR"
Key: "LEMON“

1.Shift Calculations:

1. L (11) - L (11) = 0 → A
2. X (23) - E (4) = 19 → T
3. F (5) - M (12) = 19 → T
4. O (14) - O (14) = 0 → A
5. P (15) - N (13) = 2 → C
6. V (21) - L (11) = 10 → K

A	B	C	D	E	F	G	H
0	1	2	3	4	5	6	7
I	J	K	L	M	N	O	P
8	9	10	11	12	13	14	15
Q	R	S	T	U	V	W	X
16	17	18	19	20	21	22	23
			Y	Z			
			24	25			

2.Recovered Plaintext:

"ATTACKATDAWN"

Strengths and Weaknesses of Vigenère Cipher

Strengths:

- Resists simple frequency analysis.
- More secure than monoalphabetic ciphers, especially with long keys.

Weaknesses:

- Vulnerable to **Frequency Analysis** if the key is short or reused.
- Repeating keys make it possible to detect patterns.

Breaking the Vigenère Cipher

Kasiski Examination: By analyzing repeated segments in ciphertext, one can determine the key length.

Frequency Analysis: Once the key length is known, ciphertext can be broken down into Caesar ciphers for each alphabet.

Example of Attack: Suppose we know the key length is 5. We divide the ciphertext into 5 groups and analyze each group for letter frequency.

Example in Practice

Scenario:

You are an intelligence officer needing to send a secure message to your team. The plaintext message is "MEETMEATTHEBASE" and your shared key is "SECRET".

Example in Practice

Plaintext: "MEETMEATTHEBASE“

Key: "SECRETSECRETSEC" (repeat to match length)

Encryption:

M (12) + S (18) = 30 → E
E (4) + E (4) = 8 → I
E (4) + C (2) = 6 → G
T (19) + R (17) = 36 (mod 26) → K
M (12) + E (4) = 16 → Q
E (4) + T (19) = 23 → X

A	B	C	D	E	F	G	H
0	1	2	3	4	5	6	7
I	J	K	L	M	N	O	P
8	9	10	11	12	13	14	15
Q	R	S	T	U	V	W	X
16	17	18	19	20	21	22	23
			Y	Z			
			24	25			

Resulting Ciphertext: "EIGKQXPWKRWQVO"

Practice Questions

Encrypt the following using the key "COMPUTER“

Plaintext: "HELLOCRYPTOWORLD“

Decrypt the following ciphertext using the key "KEYWORD"

Ciphertext: "QPWIGVPEZXYX"

Auto-key Vigenère Cipher

Definition:
The Auto-key Vigenère Cipher is a polyalphabetic cipher similar to the traditional Vigenère Cipher. However, instead of repeating the key, the plaintext is appended to the key after the initial key phrase.

Purpose:
The main objective is to minimize key repetition, which is a weakness in the regular Vigenère cipher, making it more resistant to frequency analysis.

Example of Auto-key Vigenère Cipher Encryption

Plaintext: "MEETMEATTHEBASE"
Key: "LEMON" (initial key) + "MEETMEATTHEBASE" (plaintext)

1.Step-by-Step Encryption:

1. First, use the key "LEMON".
2. Then, append the plaintext "MEETMEATTHEBASE" to the key after the first 5 letters.

2.Key becomes:

LEMONMEETMEATTH

Example of Auto-key Vigenère Cipher Encryption

Plaintext: "MEETMEATTHEBASE"
Key: LEMONMEETMEATTH

Encrypt the Message:

M (12) + L (11) = 23 → X
E (4) + E (4) = 8 → I
E (4) + M (12) = 16 → Q
T (19) + O (14) = 33 (mod 26) → H
M (12) + N (13) = 25 → Z

Ciphertext: "XIQHZ..."

A	B	C	D	E	F	G	H
0	1	2	3	4	5	6	7
I	J	K	L	M	N	O	P
8	9	10	11	12	13	14	15
Q	R	S	T	U	V	W	X
16	17	18	19	20	21	22	23
			Y	Z			
			24	25			

Example of Auto-key Vigenère Cipher Decryption

Ciphertext: "XIQHZ..."
Key: "LEMON" (same key, followed by recovered plaintext)

1. **Step-by-Step Decryption**:
   Use the first part of the key "LEMON".
   Decrypt the first 5 letters of the ciphertext to get the first 5 letters of the plaintext.

1. X (23) - L (11) = 12 → M
2. I (8) - E (4) = 4 → E
3. Q (16) - M (12) = 4 → E
4. H (7) - O (14) = 19 → T
5. Z (25) - N (13) = 12 → M

2. The recovered plaintext is "MEETM...". You then append this recovered plaintext to the key and proceed with the rest of the decryption.

Key Features of the Auto-key Cipher

- **More Secure than Standard Vigenère:**

  By appending plaintext to the key, it avoids the key repetition problem of the standard Vigenère cipher.

- **Key Length Matches the Plaintext Length:**

  Each letter is encrypted using a unique shift value.

- **Still Vulnerable to Known Plaintext Attacks:**

  If an attacker has access to part of the plaintext, they may reconstruct the key and break the cipher..

Practice Questions

- **Encrypt the following using the key "MATH“:**

  Plaintext: "ENCRYPTTHISMESSAGE“

- **Decrypt the following ciphertext using the key "KEY“**

  Ciphertext: "ZFTYQOKYPX"

Thank You

Question and Answers

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